

Same Chart, Different Picture

Staff-reported sparkline vividness and the Living Dashboard's uptake gap

Petra Umbile · Osei Vandermeer — School of Emergent Priorities

Introduction

The Living Dashboard assumes every viewer reads a rising sparkline the same way. Weekly engagement varies sharply by department, and little is known whether that gap reflects disengagement or a difference in whether staff can picture a trend line at all.

Objectives

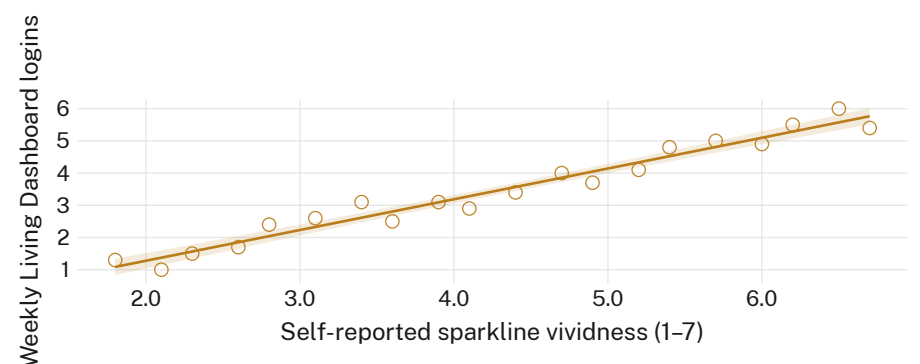
- Test whether weekly dashboard engagement tracks staff-reported visual-imagery vividness across departments
- Establish whether the gap survives controlling for role seniority and prior dashboard training
- Identify whether a non-visual trend summary narrows the gap in low-vividness departments

Materials & methods

- n = 214 staff · 9 departments · 7-point vividness self-report · 11-month login telemetry
- mixed-effects regression, department as random intercept, seniority and training as covariates
- engagement threshold (3 sessions/week) pre-registered before the telemetry pull

What we found

Self-reported sparkline vividness tracked weekly dashboard engagement across departments ($r = 0.68$, $n = 214$). The lowest-vividness department logged 61% fewer sessions than the University median, a gap prior training records did not explain.



Mean weekly Living Dashboard logins rose with self-reported sparkline vividness across 214 staff surveyed in April 2026 ($r = 0.68$); the bottom vividness tertile logged 1.6 sessions per week against a 4.1 University median.

Interpretation

The uptake gap may be a legibility failure, not a motivation one: a sparkline privileges staff who can hold a rising line in mind at a glance. Training records show no compensating effect, so redesigning the instrument looks more promising than redesigning the incentive. The association held after adjusting for seniority and tenure, suggesting the gap tracks how a trend is perceived rather than who is looking. Reported vividness clustered by department more tightly than by role, which points to a shared reading culture forming around each unit's own screens. Next: trial a text-summary companion beside the existing sparkline in the lowest-vividness department.

Further reading

1. Cleveland, W. S. & McGill, R. (1984). Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods, *Journal of the American Statistical Association*, 79(387), 531–554. doi:10.1080/01621459.1984.10478080
2. Boy, J., et al. (2014). A Principled Way of Assessing Visualization Literacy, *IEEE Transactions on Visualization and Computer Graphics*, 20(12), 1963–1972. doi:10.1109/TVCG.2014.2346984
3. Lee, S., Kim, S., & Kwon, B. C. (2017). VLAT: Development of a Visualization Literacy Assessment Test, *IEEE Transactions on Visualization and Computer Graphics*, 23(1), 551–560. doi:10.1109/TVCG.2016.2598920
4. Sarikaya, A., et al. (2019). What Do We Talk About When We Talk About Dashboards?, *IEEE Transactions on Visualization and Computer Graphics*, 25(1), 682–692. doi:10.1109/TVCG.2018.2864903
5. Galesic, M. & Garcia-Retamero, R. (2011). Graph Literacy, *Medical Decision Making*, 31(3), 444–457. doi:10.1177/0272989X10373805
6. Ziemkiewicz, C., et al. (2013). How Visualization Layout Relates to Locus of Control and Other Personality Factors, *IEEE Transactions on Visualization and Computer Graphics*, 19(7), 1109–1121. doi:10.1109/TVCG.2012.180

Vividness self-reports retained only in aggregate; login telemetry de-identified at the department level; thresholds pre-registered. Supported by a School of Continuous Improvement seed grant; we thank the staff of the participating departments.

Office of Research Outputs · slop.university · doi:10.5555/slop.qnkbze